Artificial Intelligence and Machine Learning in Chemistry

TCCM Homework: Data-Driven and ML Colour Analysis of Portuguese Wines

Assignment set by Prof. Romuald Poteau

Multiscale, Machine Learning, and QSAR Methods Applied to Biomolecules Intensive Course

Friday 29 November, Aspet

1. Project Overview

This homework is part of the **Multiscale, Machine Learning and QSAR Methods Applied to Biomolecules** module, delivered as an intensive session as part of the TCCM Masters Program. The objective is to build a machine learning pipeline in Python to classify red and white Portuguese Vinho Verde wines based solely on their physicochemical descriptors.

2. Dataset

Two datasets (one per wine colour) are available from the UCI Machine Learning Repository. The input variables (physicochemical descriptors) are:

- Fixed acidity
- Volatile acidity
- Citric acid
- Residual sugar
- Chlorides
- Free sulfur dioxide
- Total sulfur dioxide
- Density
- pH
- Sulphates
- Alcohol

The **output variable (target)** is the wine colour: red or white.

3. Bibliography

Two main references are provided:

- **Original paper:** Modeling wine preferences by data mining from physicochemical properties (2009), by Paulo Cortez, António Cerdeira, Fernando Almeida, Telmo Matos, José Reis. *Decision Support Systems* **47**: 547-553.
- **Wine Project GitHub repository** (Wine Quality website): authors explored several classifiers including K-Nearest Neighbors, Decision Tree, Random Forest, Random Forest Regressor, Stochastic Gradient Descent, Support Vector Classification, Linear SVC, AdaBoost, Gradient Boost, and XGBoost, obtaining 69%-95% performance. The highest performers were Random Forest Regression, Random Forest, and XGBoost.

Some useful code snippets are available on the Wine Quality website and on the pyPhysChem repository (e.g. automatic one-hot encoding in DS4B-Appendix.ipynb and tools4PytChem.py).

4. Mandatory Implementation

The following requirements must be satisfied:

- Code must be written in **Python**, in a **Jupyter notebook**, using primarily **Keras** and **TensorFlow**; scikit-learn tools may also be used where appropriate.
- The notebook must be commented using both **Markdown cells** and **Python inline/block comments**. No separate report is required - the notebook itself serves as the deliverable.
- Particular attention should be paid to **notebook organisation**: use headers, ordered and unordered lists, tables, mathematical formulae, hyperlinks, and images where relevant.
- Data must be **cleaned and prepared**. A **correlation matrix** must be plotted separately for red and white wines. It may also be useful to build a combined dataset with wine colour as a 12th descriptor.
- Both a **simple ANN** and a **deep learning approach** should be explored. A **confusion matrix** must be produced for each architecture tested.
- The numerical stability of the best model must be assessed using **5-fold cross-validation**.
- The **SHAP explanation method** must be applied to interpret feature contributions to model predictions. Discuss whether the Shapley values are consistent with the exploratory data analysis.

5. Recommended Libraries and Resources

The following tools and reference notebooks (from the **pyPhysChem** repository) are recommended:

- **pandas** - data management (see iris1 notebook for basic applications)
- **keras** - ANN definition and training (see CO2 solubility in ILs notebook)
- **scikit-learn** - logistic regression (see iris3 notebook for Iris species identification)
- **matplotlib** and **seaborn** - graphical analysis and visualisation
- **shap** - XAI / explainability (see iris4 notebook)
- *Note: the rdkit library is not needed, as wines differ only in the concentration of chemical compounds, not in their molecular nature.*

6. Prerequisites

This module follows a **flipped classroom model**. Before the hands-on session, participants are required to have read, run, and understood the following three notebooks from the pyPhysChem repository:

- Reading and analysis of the Iris database with the pandas library
- Statistics and regression
- Basics of Artificial Neural Networks (ANN) for supervised learning

All teaching materials and installation instructions are available at: https://github.com/rpoteau/pyPhysChem